Activity 6

Take a triangle like this

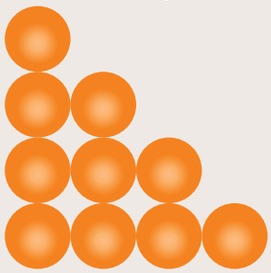


Fig.2.16 $(1+2+3+4)$

Make another triangle like this.

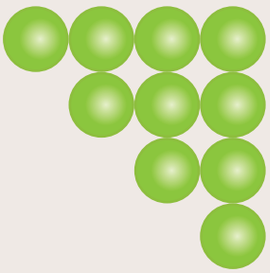


Fig.2.17 $(4+3+2+1)$

Join the second triangle with the first to get

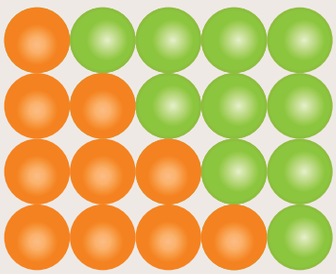


Fig.2.18

Thus, two copies of $1+2+3+4$ provide a rectangle of size 4×5.

We can write in numbers, what we did with pictures.

Let us write, $(4+3+2+1)+(1+2+3+4)=4\times5$

$$2(1+2+3+4)=4\times5$$

Therefore, $1+2+3+4=\dfrac{4\times5}{2}=10$

In a similar, fashion, try to find the sum of first 5 natural numbers. Can you relate these answers to any of the known formula?

DO YOU KNOW?

1. The sum of first n natural numbers are also called Triangular Numbers because they form triangle shapes.
2. The sum of squares of first n natural numbers are also called Square Pyramidal Numbers because they form pyramid shapes with square base.

Thinking Corner

1. How many squares are there in a standard chess board?
2. How many rectangles are there in a standard chess board?

Here is a summary of list of some useful summation formulae which we discussed. These formulae are used in solving summation problems with finite terms.

$$\sum_{k=1}^{n} k = 1+2+3+\cdots+n = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^{n} (2k-1) = 1+2+3+\cdots+(2n-1) = n^2$$

$$\sum_{k=1}^{n} k^2 = 1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^{n} k^3 = 1^3+2^3+3^3+\cdots+n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

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